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"""
Project 1: The To-Do List
DecodeLabs - Python Programming Industrial
Training Kit

Goal: Build a program where users can add
tasks to a list and view them.
Key Skill: Lists (append & print loops)

Bonus: Persistence — tasks are saved to a
JSON file on disk (DISK, not
just volatile RAM), so your list survives
between runs of the program.
"""

import json
import os

DATA_FILE = "tasks.json"


def load_tasks():
    """Read tasks from disk into memory
(RAM) on startup."""
    if os.path.exists(DATA_FILE):
        try:
            with open(DATA_FILE, "r") as f:
                return json.load(f)
        except (json.JSONDecodeError,
IOError):
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            print("Couldn't read saved
tasks — starting with an empty list.")
    return []


def save_tasks(tasks):
    """Write the current in-memory list to
disk (serialization)."""
    with open(DATA_FILE, "w") as f:
        json.dump(tasks, f, indent=2)


def show_menu():
    print("\n===== TO-DO LIST MENU =====")
    print("1. Add a task")
    print("2. View all tasks")
    print("3. Remove a task")
    print("4. Exit")


def add_task(tasks):
    task = input("Enter the task to add:
").strip()
    if task:
        tasks.append(task)  # store the new
item in the list
        save_tasks(tasks)   # persist
immediately to disk
        print(f"'{task}' has been added to
your to-do list.")
    else:
        print("Task cannot be empty.")
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def view_tasks(tasks):
    if not tasks:
        print("Your to-do list is empty.")
        return

    print("\n--- YOUR TASKS ---")
    for index, task in enumerate(tasks,
start=1):  # enumerate gives (index, value)
        print(f"{index}. {task}")


def remove_task(tasks):
    if not tasks:
        print("Nothing to remove — your
list is empty.")
        return

    view_tasks(tasks)
    try:
        choice = int(input("Enter the
number of the task to remove: "))
        if 1 <= choice <= len(tasks):
            removed = tasks.pop(choice - 1)
            save_tasks(tasks)  # persist
immediately to disk
            print(f"Removed: '{removed}'")
        else:
            print("Invalid task number.")
    except ValueError:
        print("Please enter a valid
number.")


def main():
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    my_tasks = load_tasks()  # load from
disk into memory on startup

    while True:
        show_menu()
        choice = input("Choose an option
(1-4): ").strip()

        if choice == "1":
            add_task(my_tasks)
        elif choice == "2":
            view_tasks(my_tasks)
        elif choice == "3":
            remove_task(my_tasks)
        elif choice == "4":
            print(f"Goodbye! Your tasks are
saved in '{DATA_FILE}'.")
            break
        else:
            print("Invalid choice. Please
select 1, 2, 3, or 4.")


if __name__ == "__main__":
    main()
```